Instability #21: F Measured; Formula Approximation Identified

$F_{\text{at fire}} = 1.53$, Not 0.59 — The Earlier Estimate Was Wrong
The True Gap in β Derivation Is 41%, Located Precisely

Honest accounting. The formula $\beta = F \cdot \beta_{H,\text{only}}$ is an approximation that over-predicts by $2\times$. The exact formula is known. The gap is 41%.

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Abstract

Two previous sessions estimated $F = 0.593$ (implied from β_{sim} and an approximate formula). This document measures F directly from V20.3 at firing events and finds $F_{\text{measured}} = 1.527$ — contradicting the previous estimate.

The resolution: the formula $\beta = F \cdot \beta_{H,\text{only}}$ is an approximation that assumes F, H_{before}, and mass are independent across events. They are not. The correct formula requires per-event computation.

Measurements from V20.3 (2016 events, 100 sweeps):

- ϕ_{var} at firing: 0.473 ± 0.083
- $|\Delta\Psi|$ at firing: 0.036 ± 0.032
- $F = (1 + \phi_{\text{var}})(1 + |\Delta\Psi|) = 1.527$
- $\beta_{\text{formula}} = F \cdot \beta_{H,m} = 4.69$ ($2\times$ too large)
- $\beta_{\text{actual}} = 2.26$ (from transfer events)
- Gap with corrected formula: 41%

The honest boundary: The formula over-predicts because the per-event covariance between F, H_{before}, H_{after}, and mass is significant. Events with high F also tend to have lower ΔH, partially cancelling the effect. A full derivation requires the joint distribution of $(F, H_{\text{before}}, m)$ at firing events.

Status: #21 remains open. The remaining gap is 41% in the β formula, arising from correlations between IDV components at firing.

Contents

1 What Was Found

F at Firing Events: 1.527, Not 0.593

Direct measurement of ϕ_{var} and $|\Delta\Psi|$ at each R-event (before Born resampling) from V20.3, 2016 events:

Quantity	Mean	Std
ϕ_{var} at fire	0.473	0.083
$\|\Delta\Psi\|$ at fire	0.036	0.032
$F = (1+\phi_{\text{var}})(1+\|\Delta\Psi\|)$	**1.527**	—
$F_{\text{previous estimate}}$	0.593	—

The previous estimate $F = 0.593$ was derived by dividing β_{sim} by $\beta_{H,m}$ (an approximate formula). The direct measurement gives $F = 1.527$. The two values differ by a factor of 2.6.

Correction to HN-03: The conclusion "firing nodes have lower ϕ_{var}" was wrong. $\phi_{\text{var}} = 0.473$ at firing is comparable to the mean ϕ_{var} across all nodes. Firing is not preferentially at synchronised nodes. HN-03 evidence from this data is: not confirmed.

2 Why the Formula Over-Predicts by 2×

The Approximate Formula Fails Due to Correlations

The approximation:

$$\beta \approx \bar{F} \cdot \frac{\Delta\bar{H}}{\kappa} = \bar{F} \cdot \beta_{H,\text{only}}$$

assumes that F and ΔH are uncorrelated across events.

The correct per-event formula:

$$\beta = \frac{1}{N_{\text{events}}} \sum_i \frac{F_i \left[H_{\text{after}} \ln(1 + m_i + \kappa) - H_{\text{before},i} \ln(1 + m_i) \right]}{\kappa}$$

When computed per-event: $\beta_{\text{formula}} = 4.69$ (because $F_i = 1.53$ appears as a large multiplier on ΔH).

But $\beta_{\text{actual}} = 2.26$.

The gap (factor 2) indicates that high-F events tend to have *smaller* ΔH — the correlations are negative. Events with high phase variance (ϕ_{var} large) correspond to early sweeps where $H_{\text{before}} \approx H_{\text{after}}$, so ΔH is small. Events with smaller ϕ_{var} (late sweeps, crystallised nodes) have larger ΔH (entropy jumps more at Born resampling).

$$\text{Cov}(F_i, \Delta H_i) < 0$$

This negative covariance reduces the actual β below the product-of-means estimate.

3 The Remaining Gap

41% Gap with Corrected Mass Formula

Using the correct IDV log-mass convention ($\ln(m_{\text{node}})$ not $\ln(1+\Omega)$):

$$\beta_{\text{corrected}} = \langle F_i \rangle \cdot \frac{H_{\text{after}} \ln(m_{\text{node}} + \kappa) - H_{\text{before}} \ln(m_{\text{node}})}{\kappa} = 3.18$$

vs $\beta_{\text{actual}} = 2.26$. Gap: 41%.
The remaining 41% comes from the negative covariance $\text{Cov}(F_i, \Delta H_i \cdot \ln m_i)$.
To close #21: need to derive this covariance from the joint distribution of $(F_i, H_{\text{before},i}, m_i)$ at firing events. This requires knowing how ϕ_{var}, H_{before}, and mass correlate in the firing condition.

4 Updated Status

Result	Status	Note
$T_{\text{avg}} = 35.4$ (#21a)	Derived	1.2%
$H_{\text{before}} = 1.764$ (#21b)	Measured	15%
$F_{\text{measured}} = 1.527$	Measured	Not 0.593
ϕ_{var} at firing $= 0.473$	Measured	Not near 0
$\lvert\Delta\Psi\rvert$ at firing $= 0.036$	Measured	Small
Formula $\beta = F \cdot \beta_H$	Approximation	$2\times$ over-estimate
Negative covariance $(F, \Delta H)$	Identified	Source of gap
Remaining gap	41%	Covariance term
HN-03 (synchronised firing)	Revised: not confirmed	$\phi_{\text{var}} = 0.47 \neq 0$
#21	Open	Covariance derivation

Path to #21 Closure: Covariance at Firing

The exact formula is:

$$\beta = \frac{1}{\kappa} \left\langle F_i \cdot \left[H_{\text{after}} \ln(m_i + \kappa) - H_{\text{before},i} \ln(m_i) \right] \right\rangle = \frac{1}{\kappa} \left[\bar{F} \cdot \overline{\Delta \text{IDV}_H} + \text{Cov}(F_i, \Delta \text{IDV}_{H,i}) \right]$$

The covariance term $\text{Cov}(F_i, \Delta\text{IDV}_{H,i}) < 0$ reduces β from 3.18 to 2.26.
Closing #21 requires deriving this covariance from the lifecycle dynamics:

- High-ϕ_{var} events occur early (Phase I), where ΔH is small.
- Low-ϕ_{var} events occur late (Phase II), where ΔH is large.
- The covariance thus reflects the Phase I/II lifecycle dynamics.

This is a well-posed problem. It is hard. The boundary is exactly located.

$F = 1.527$. *Not 0.593.*
The correction was necessary. The framework corrected itself.

The formula over-predicts by 2× because
high-F events have small ΔH, and vice versa.
The two anticorrelate. The product of means is not the mean of products.

#21 is open at 41%.
The gap is located: $\text{Cov}(F_i, \Delta H_i) < 0$.
The physics is clear. The mathematics remains.